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Inventor(s)	Oh; Taewoong

Electronic parking brake system and method of controlling the same

Abstract

Disclosed herein is an electronic parking brake system including an electronic parking brake that provides a braking force to a vehicle, and a controller that is electrically connected to the electronic parking brake and controls the electronic parking brake, wherein the controller performs theft prevention control by the electronic parking brake according to a driving state of a stolen vehicle, based on reception of a theft prevention request.

Inventors: **Oh; Taewoong (Suwon-si, KR)**

Applicant: **HL MANDO CORPORATION** (Pyeongtaek-si, KR)

Family ID: **87933104**

Assignee: **HL MANDO CORPORATION (Pyeongtaek-si, KR)**

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Primary Examiner: Solis; Erick R

Attorney, Agent or Firm: Hauptman Ham, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of Korean Patent Application No. 10-2022-0030151, filed on Mar. 10, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

(2) Embodiments of the present disclosure relate to an electronic parking brake system having an electronic parking brake operated by a motor, and a method of controlling the same.

2. Description of the Related Art

(3) In general, as a key-insertion and rotation-type start-up method is changed to a smart key, a digital key, or the like, in which a physical key disappears, technologies for preventing stealing of a vehicle have been introduced in various manners.

(4) An electronic parking brake system increases a torque generated from a motor through a reducer to generate a braking force required for parking by a mechanical structure device inside a caliper.

(5) In recent years, technologies for stopping the vehicle by using the electronic parking brake system when the vehicle is stolen have been developed.
(6) However, in the related art, when the vehicle is stolen, a constant braking force is generated regardless of a driving state of the stolen vehicle. Accordingly, when a relatively large braking force is generated in a driving condition of the stolen vehicle, the stolen vehicle is suddenly stopped, and thus an accident may occur. Further, when a relatively small braking force is generated in the driving condition of the stolen vehicle, the stolen vehicle cannot be stopped. In addition, considering a case in which a vehicle thief additionally manipulates an accelerator pedal to prevent the stolen vehicle from stopping, it is more difficult to safely and reliably stop the stolen vehicle using only a manner of generating a constant braking force.
RELATED ART DOCUMENT
Patent Document
(7) (Patent Document 1) Japanese Patent Application Publication No. 2002-160603 (published on Jun. 4, 2002)
SUMMARY
(8) Therefore, it is an aspect of the present disclosure to provide an electronic parking brake system which safely and reliably stops a stolen vehicle by generating a braking force for stop control according to a driving state of the stolen vehicle when a theft prevention request is received, and a method of controlling the same.
(9) Additional aspects of the disclosure will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the disclosure.
(10) In accordance with one aspect of the present disclosure, an electronic parking brake system includes an electronic parking brake that provides a braking force to a vehicle, and a controller that is electrically connected to the electronic parking brake and controls the electronic parking brake. The controller performs theft prevention control with the electronic parking brake according to a driving state of a stolen vehicle, based on reception of a theft prevention request.
(11) The controller may perform deceleration control in which a deceleration of the stolen vehicle is maintained at a target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake, based on driving of the stolen vehicle.
(12) The controller may forcibly engage the electronic parking brake, based on stopping of the stolen vehicle by the deceleration control.
(13) The controller may forcibly engage the electronic parking brake while the stolen vehicle is stopping.
(14) While performing the theft prevention control, the controller may ignore a parking release request from an electronic parking brake (EPB) switch and prohibit an operation stop of the electronic parking brake or a release of the electronic parking brake.
(15) The controller may perform deceleration control in which a deceleration of the stolen vehicle is maintained at a target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake, based on a speed of the stolen vehicle, which is higher than a preset speed, and may forcibly engage the electronic parking brake, based on the speed of the stolen vehicle, which is lower than the preset speed.
(16) The controller may start the theft prevention control at a time point at which an engagement of the electronic parking brake is requested.
(17) The controller may start deceleration control in which a deceleration of the stolen vehicle is maintained at a target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake at a time point at which a speed of the stolen vehicle is lower than a preset speed.
(18) The controller may output theft prevention control completion so that a vehicle owner identifies completion of the theft prevention control, based on the completion of the theft prevention control.
(19) The controller may release the theft prevention control, based on reception of a theft

prevention release request.
(20) The electronic parking brake system may further include a warner that guides information to a driver. The controller may output, to the warner, information notifying that the theft prevention control is released, based on the reception of the theft prevention release request.
(21) The controller may output, to the warner, information notifying that a release of the electronic parking brake is permitted, based on forcibly engaging of the electronic parking brake by the theft prevention control.
(22) In accordance with another aspect of the present disclosure, a method of controlling an electronic parking brake system that controls an electronic parking brake that provides a braking force to a vehicle includes determining whether a theft prevention request is received, determining a driving state of a stolen vehicle, based on the reception of the theft prevention request, and performing theft prevention control by with the electronic parking brake according to the driving state of the stolen vehicle.
(23) The performing of the theft prevention control may include performing deceleration control in which a deceleration of the stolen vehicle is maintained at a target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake, based on driving of the stolen vehicle, and forcibly engaging the electronic parking brake, based on stopping of the stolen vehicle by the deceleration control.
(24) The performing of the theft prevention control may include ignoring a parking release request from an electronic parking brake (EPB) switch while the theft prevention control is performed and prohibiting an operation stop of the electronic parking brake or a release of the electronic parking brake.
(25) The performing of the theft prevention control may include starting the theft prevention control at a time point at which engagement of the electronic barking brake is requested or at a time point at which a speed of the stolen vehicle is lower than a preset speed.
(26) The method may further include outputting theft prevention control completion so that a vehicle owner identifies completion of the theft prevention control, based on the completion of the theft prevention control.
(27) The method may further include releasing the theft prevention control, based on reception of a theft prevention release request, and outputting information notifying that a release of the electronic parking brake is permitted, based on forcible engaging of the electronic parking brake by the theft prevention control.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) These and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings of which:
(2) FIG. **1** is a diagram of an electronic parking brake included in an electronic parking brake system according to an embodiment;
(3) FIG. **2** is a control block diagram of the electronic parking brake system according to the embodiment;
(4) FIG. **3** is a control flowchart of the electronic parking brake system according to the embodiment;
(5) FIG. **4** is a sequence diagram for describing a process of performing theft prevention control according to a theft prevention request from a vehicle owner in the electronic parking brake system according to the embodiment; and
(6) FIG. **5** is a sequence diagram for describing a process of performing theft prevention release

according to a theft prevention release request from a vehicle owner in the electronic parking brake system according to the embodiment.

DETAILED DESCRIPTION

(7) Throughout the specification, the same reference numerals refer to the same components. The present specification does not describe all components of the embodiments, and general content in the technical field to which the present disclosure pertains or duplicated content between the embodiments will be omitted. The terms “unit,” “module,” “member,” and “block” used in the specification may be implemented as software or hardware, and according to embodiments, a plurality of “units,” “modules,” “members,” and “blocks” may be implemented as a single component or one “unit,” “module,” “member,” or “block” may include a plurality of components.

(8) Throughout the specification, when it is described that a first component is “connected” to a second component, this includes not only a case in which the first component is directly connected to the second component but also a case in which the first component is indirectly connected to the second component, and the indirect connection includes connection through a wireless communication network.

(9) Further, when a part “includes” a component, this does not mean that other components are excluded but in fact other components may be further included unless otherwise stated.

(10) Throughout the specification, when a first member is located “on” a second member, this includes not only a case in which the first member is in contact with the second member but also a case in which a third member is present between the two members.

(11) Terms such as “first” and “second” are used to distinguish one component from another component, and components are not limited by the above-described terms. Singular expressions include plural expressions unless clearly otherwise indicated in the context.

(12) In each of operations, an identification code is used for convenience of description and does not describe a sequence of the operations, and the operations may be performed in a different order from the specified order unless the context clearly states a specific order.

(13) FIG. **1** is a diagram of an electronic parking brake included in an electronic parking brake system according to an embodiment.

(14) Referring to FIG. **1**, an electronic parking brake **10** may include a carrier **110** in which a pair of pad plates **111** and **112** are installed to move forward or rearward to press a brake disc **100** rotating together with a wheel of a vehicle, a caliper housing **120** provided with a cylinder **123** slidably installed in the carrier **110** and installed so that a piston **121** moves forward or rearward under a braking hydraulic pressure, a power converter **130** configured to press the piston **121**, and a motor actuator **140** configured to transmit a rotational force to the power converter **130** using a motor M.

(15) The pair of pad plates **111** and **112** may be divided into an inner pad plate **111** disposed in contact with the piston **121** and an outer pad plate **112** disposed in contact with a finger portion **122** of the caliper housing **120**. The pair of pad plates **111** and **112** may be installed on the carrier **110** fixed to a vehicle body to move forward or rearward toward both sides of the brake disc **100**. Further, a brake pad **113** may be attached to one surface of each of the pad plates **111** and **112** facing the brake disc **100**.

(16) The caliper housing **120** may be slidably installed in the carrier **110**. In more detail, the caliper housing **120** may include the cylinder **123** on a rear side thereof in which the power converter **130** is installed and the piston **121** is embedded to move forward or rearward and the finger portion **122** formed to be bent on a front side thereof in a downward direction to operate the outer pad plate **112**. The finger portion **122** and the cylinder **123** may be integrally formed.

(17) The piston **121** may be provided in a cylindrical shape having a cup-shaped interior and slidably inserted into the cylinder **123**. The piston **121** may press the inner pad plate **111** toward the brake disc **100** with an axial force of the power converter **130** configured to receive the rotational force of the motor actuator **140**. Accordingly, when the axial force of the power converter **130** is

applied, the piston **121** moves forward toward the inner pad plate **111** to press the inner pad plate **111**, the caliper housing **120** operates in a direction opposite to the piston **121** due to a reaction force so that the finger portion **122** presses the outer pad plate **112** toward the brake disc **100**, and thus braking can be performed.
(18) The power converter **130** may serve to press the piston **121** toward the inner pad plate **111** by receiving the rotational force from the motor actuator **140**.
(19) The power converter **130** may include a nut member **131** that is installed to be disposed inside the piston **121** and is in contact with the piston **121** and a spindle member **135** screw-coupled to the nut member **131**.
(20) The nut member **131** may be disposed inside the piston **121** in a rotation-restricted state and screw-coupled to the spindle member **135**.
(21) The nut member **131** may include a head portion **132** provided in contact with the piston **121** and a coupling portion **133** extending from the head portion **132** and having a female thread formed on an inner circumferential surface thereof to be screw-coupled to the spindle member **135**.
(22) The nut member **131** may serve to press the piston **121** or release the pressing of the piston **121** while moving in a forward direction or a rearward direction according to a rotation direction of the spindle member **135**. In this case, the forward direction may be a movement direction in which the nut member **131** approaches the piston **121**. The rearward direction may be a movement direction in which the nut member **131** moves away from the piston **121**. In addition, the forward direction may be a movement direction in which the piston **121** approaches the brake pad **113**. The rearward direction may be a movement direction in which the piston **121** moves away from the brake pad **113**.
(23) The spindle member **135** may include a shaft **136** passing through a rear portion of the caliper housing **120** and rotating by receiving the rotational force of the motor actuator **140**, and a flange portion **137** extending radially from the shaft **136**. One side of the shaft **136** may be rotatably installed to pass through a rear side of the cylinder **123**, and the other side thereof may be disposed inside the piston **121**. In this case, the one side of the shaft **136** passing through the cylinder **123** may be connected to an output shaft of a reducer **142** to receive the rotational force of the motor actuator **140**.
(24) The motor actuator **140** may include a motor **141** and the reducer **142**.
(25) The motor **141** may rotate the spindle member **135** to move the nut member **131** forward or rearward to press the piston **121** or release the pressing of the piston **121**.
(26) The reducer **142** may be provided between an output side of the motor **141** and the spindle member **135**.
(27) Due to the above configuration, the electronic parking brake **10** may press the piston **121** by moving the nut member **131** by rotating the spindle member **135** in one direction using the motor actuator **140** when parking is performed. The piston **121** pressed by the movement of the nut member **131** may press the inner pad plate **111** to bring the brake pad **113** into close contact with the brake disc **100**, thereby generating an engaging force.
(28) Further, the electronic parking brake **10** rotates the spindle member **135** in an opposite direction using the motor actuator **140** when the parking is released so that the nut member **131** pressed by the piston **121** may move rearward. The pressing on the piston **121** may be released by the rearward movement of the nut member **131**. By releasing the pressing on the piston **121**, the brake pad **113** is spaced apart from the brake disc **100**, and thus the generated engaging force may be released.
(29) FIG. **2** is a control block diagram of the electronic parking brake system according to the embodiment.
(30) Referring to FIG. **2**, the electronic parking brake system may include the electronic parking brake **10**, an electronic parking brake (EPB) switch **20**, a controller **30**, a current sensor **40**, and a warner **50**.

(31) The electronic parking brake **10** may provide a parking braking force to the brake disc **100** rotating together with wheels, for example, left and right rear wheels, of the vehicle.
(32) The electronic parking brake **10** generates the parking braking force for each rear wheel. The electronic parking brake **10** is controlled by the electrically connected controller **30**.
(33) The electronic parking brake **10** is operated by the motor **141** to generate the parking braking force. The electronic parking brake **10** may operate the motor **141** to press the brake pad **113** inside the caliper housing **120** on the left and right rear wheels to generate the parking braking force.
(34) The EPB switch **20** is a switch configured to receive an intention of a driver to operate the electronic parking brake **10** and may be provided adjacent to a driver's seat of the vehicle.
(35) The EPB switch **20** is provided to be turned on or off by the driver.
(36) The EPB switch **20** transmits, to the controller **30**, a signal corresponding to a braking operation command (fastening command) when the EPB switch **20** is turned on and a signal corresponding to a parking release command (unfastening command) when the EPB switch **20** is turned off.
(37) The current sensor **40** may be provided to detect a current flowing through the motor **141** of the electronic parking brake **10**. The current sensor **40** may detect a motor current flowing through the motor **141** using a shunt resistor or a Hall sensor. Various methods that can detect the motor current in addition to the shunt resistor and the Hall sensor may be applied to the current sensor **40**.
(38) The warner **50** may be a warning light, a cluster, or a combination thereof. The warner **50** may warn the driver through video, audio or both.
(39) The warner **50** may output a message or a warning sound indicating that the theft prevention control is operating to warn a vehicle thief.
(40) The controller **30** may be called an electronic control unit (ECU).
(41) The controller **30** may include a processor **31** and a memory **32**.
(42) The processor **31** may include a digital signal processor configured to process information related to the operation of the electronic parking brake **10** and a micro control unit (MCU) configured to generate a motor driving signal for the parking operation, the parking release, or the theft prevention control of the electronic parking brake **10**.
(43) The memory **32** may store a program and/or data for processing the information related to the operation of the electronic parking brake **10** and a program and/or data for generating the motor driving signal for the parking operation, the parking release or the theft prevention control of the electronic parking brake **10** by the processor **31**.
(44) The memory **32** may temporarily store the information related to the operation of the electronic parking brake **10** and temporarily store a processing result of the information related to the operation of the electronic parking brake **10** by the processor **31**.
(45) The memory **32** may include not only a volatile memory such as a static random access memory (S-RAM) or a dynamic RAM (D-RAM) but also a non-volatile memory such as a flash memory, a read only memory (ROM), or an erasable programmable ROM (EPROM).
(46) The controller **30** receives power from a battery **60**.
(47) The controller **30** supplies the power provided from the battery **60** to the motor **141** of the electronic parking brake **10** to drive the motor **141** and generate the parking braking force. When the motor **141** of the electronic parking brake **10** is driven to generate or release the parking braking force, the controller **30** may supply or block the power to or from the motor M and **141**.
(48) The controller **30** may drive the motor **141** of the electronic parking brake **10**. The controller **30** may drive the motor **141** in a forward rotation direction or a rearward rotation direction. For example, the controller **30** may include an H-bridge circuit including a plurality of power switching elements to drive the motor **141** in the forward rotation direction or the rearward rotation direction.
(49) The controller **30** may communicate with various systems mounted on the vehicle through a network bus. Ethernet, media oriented systems transport (MOST), FlexRay, a controller area network (CAN), a local interconnect network (LIN), and the like may be used as the network bus.

(50) The controller **30** may communicate with various systems mounted on the vehicle to receive information such as a wheel speed and the theft prevention request. The wheel speed may include speeds of four wheels of the vehicle. The theft prevention request is a signal generated when vehicle theft occurs. The theft prevention request may be generated by the vehicle itself and transmitted to the controller **30** on the basis of determination by the vehicle that the vehicle has been stolen. Further, when a vehicle owner manipulates a mobile phone to transmit the theft prevention request, the controller **30** may directly receive the theft prevention request or the theft prevention request may be provided from another device of the vehicle receiving the theft prevention request.
(51) The controller **30** may perform a parking operation mode or a parking release mode according to an operation signal of the EPB switch **20** or an operation signal generated by a program related to the operation of the electronic parking brake **10**.
(52) In the parking operation mode, the controller **30** determines whether a current motor current reaches a target current and terminates the parking operation control on the basis of the reaching the target current.
(53) Meanwhile, when the theft prevention request is received, the controller **30** may determine that vehicle theft has occurred and perform a theft prevention mode for forcibly stopping the stolen vehicle.
(54) In the theft prevention mode, different theft prevention controls are performed according to a driving state of the stolen vehicle.
(55) In the theft prevention mode, when the stolen vehicle is driving, the electronic parking brake **10** is operated to generate a braking force, and thus the stolen vehicle is decelerated at a target deceleration until the stolen vehicle stops.
(56) In the theft prevention mode, when the stolen vehicle is stopped, the electronic parking brake **10** is forcibly engaged to prevent the stolen vehicle from moving.
(57) When the theft prevention mode is operated, the controller **30** may output, through the warner **50**, a message or a warning sound informing that the theft prevention control is operating, to warn the vehicle thief that the theft prevention control is operating.
(58) In the theft prevention mode, when the stolen vehicle is driving, the controller **30** generates the braking force from the electronic parking brake **10** until the stolen vehicle is stopped, decelerates the stolen vehicle at the target deceleration, and thus safely and securely stops the stolen vehicle.
(59) In the theft prevention mode, when the stolen vehicle is stopped, the controller **30** forcibly engages the electronic parking brake **10** to prevent the stolen vehicle from moving.
(60) While the theft prevention mode is performed, even when the parking release command is input by the EPB switch **20**, the controller **30** ignores the parking release command and prohibits the parking release.
(61) Upon receiving a theft prevention release request for releasing the theft prevention control, the controller **30** releases the theft prevention mode.
(62) When the theft prevention mode is released, the controller **30** outputs a guide message or a sound according to the release of the theft prevention mode to notify that the theft prevention mode is released, and to guide a method for returning the vehicle to a state prior to the theft prevention mode.
(63) When the theft prevention mode is released, the controller **30** no longer decelerates the vehicle at the target deceleration while the vehicle is driving. Further, when the vehicle is stopped and the driver operates the EPB switch **20**, the parking release is performed to release the forcibly engaging of the electronic parking brake **10**.
(64) FIG. **3** is a control flowchart of the electronic parking brake system according to the embodiment.
(65) Referring to FIG. **3**, first, the controller **30** determines whether the theft prevention request is received (**200**).

(66) When the theft prevention request is received (**200**, Yes), the controller **30** performs the theft prevention mode (**202**).
(67) The controller **30** notifies the vehicle thief that the theft prevention control is operating through the warner **50** (**204**).
(68) The controller **30** detects the wheel speed of the stolen vehicle (**206**). The controller **30** may directly detect the wheel speed from a wheel speed sensor and also receive the wheel speed from another system in the vehicle.
(69) The controller **30** determines whether the vehicle speed is greater than or equal to a preset speed Vmin according to the wheel speed (**208**).
(70) When the vehicle speed is greater than or equal to the preset speed Vmin (**208**, Yes), the controller **30** determines the target deceleration for stopping the stolen vehicle (**210**). The target deceleration may be a preset deceleration for stopping the vehicle.
(71) The controller **30** operates the electronic parking brake **10** to perform deceleration control according to the target deceleration (**212**). The controller **30** performs the deceleration control by engaging the electronic parking brake **10** to generate the braking force and by adjusting the braking force generated by the electronic parking brake **10** so that a current vehicle deceleration reaches the target deceleration. The controller **30** performs the deceleration control according to the target deceleration until the vehicle speed is less than the preset speed Vmin (**212**).
(72) Meanwhile, when the vehicle speed is less than the preset speed Vmin, the controller **30** performs the parking operation to engage the electronic parking brake **10** (**214**). During the parking operation, the controller **30** drives the motor **141** to rotate the spindle member **135** in one direction to move the nut member **131** to press the piston **121**. The piston **121** pressed by the movement of the nut member **131** presses the inner pad plate **111** to bring the brake pad **113** into close contact with the brake disc **100**, thereby generating the engaging force.
(73) After performing the parking operation, the controller **30** determines that the parking operation is completed (**216**). The controller **30** may detect the current of the motor **141** through the current sensor **40** during the parking operation and determine that the parking operation is completed when the detected motor current reaches the target current.
(74) When the parking operation is completed (**216**, Yes), the controller **30** notifies of the completion of the theft prevention control (**218**). The controller **30** transmits the completion of the theft prevention control to a device in the vehicle that can notify the vehicle owner of the completion, to notify the vehicle owner that the theft prevention control is completed.
(75) FIG. **4** is a sequence diagram for describing a process of performing theft prevention control according to a theft prevention request from a vehicle owner in the electronic parking brake system according to the embodiment.
(76) FIG. **4** illustrates information exchange between an electronic parking brake system **300** provided in the vehicle, a vehicle system **310** connected to the electronic parking brake system **300** in the vehicle through a network, and the vehicle owner having a user terminal such as a mobile phone, a tablet personal computer (PC), a laptop computer, and a desktop computer, which may directly or indirectly communicate with the vehicle system **310** in a remote manner. The vehicle system **310** may include a communication device that may directly or indirectly communicate with a user terminal of an external vehicle owner **320** in a remote manner and an electronic stability control (ESC) device connected to the communication device through a network and connected to the electronic parking brake system **300** electrically or through the network. The ESC device is a device that receives wheel speed information from the wheel speed sensor provided on a wheel side, instantaneously and independently controls four wheels to maintain driving stability when the vehicle is about to slide, and thus improves the driving stability.
(77) First, when the theft of the vehicle is recognized, the vehicle owner **320** transmits the theft prevention request to the vehicle system **310** of the stolen vehicle using the user terminal. In this case, the vehicle system **310** may detect the vehicle theft by itself and transmit a message

informing that the vehicle theft has occurred to the mobile device of the vehicle owner **320**.
(78) When the vehicle system **310** receives the theft prevention request from the vehicle owner **320**, the vehicle system **310** transmits the theft prevention request to the electronic parking brake system **300** (**402**).
(79) When the electronic parking brake system **300** receives the theft prevention request from the vehicle system **310**, the electronic parking brake system **300** performs the theft prevention control of forcibly stopping the stolen vehicle using the electronic parking brake **10** when the stolen vehicle is driving and forcibly engaging the electronic parking brake **10** when the stolen vehicle is stopped (**404**). During the theft prevention control, the vehicle thief is notified that the theft prevention control is operating (**404***a*). The wheel speed of the stolen vehicle is detected (**404***b*). The deceleration control in which the electronic parking brake **10** is operated to generate the braking force and the deceleration of the stolen vehicle is controlled to the target deceleration until the stolen vehicle is stopped by varying the generated braking force is performed when the stolen vehicle is driving. When the stolen vehicle is stopped, the electronic parking brake **10** is forcibly engaged so that the stolen vehicle does not move (**404***d*). While the theft prevention control is performed, a parking release request from the EPB switch **20** is ignored, and stopping of the operation of the electronic parking brake **10** is prohibited or a release of the electronic parking brake **10** is prohibited.
(80) When the theft prevention control is completed, the electronic parking brake system **300** transmits a theft prevention control completion notification to the vehicle system **310** (**406**).
(81) When the theft prevention control completion notification is received from the electronic parking brake system **300**, the vehicle system **310** provides the vehicle owner **320** with the theft prevention control completion notification (**408**).
(82) Meanwhile, instead of starting the theft prevention control immediately after a time point when the theft prevention request is received from the vehicle system **310**, the electronic parking brake system **300** may start the theft prevention control at a time point when the vehicle thief manipulates the EPB switch **20** to request the parking operation or a specific parking operation condition (for example, a start-up off operation, an auto hold operation, or the like) is satisfied, and thus the parking operation is performed.
(83) Further, the electronic parking brake system **300** may start the theft prevention contact immediately after the time point when the theft prevention request is received from the vehicle system **310** and may start the deceleration control of the theft prevention control at a time point when a speed of the stolen vehicle is lower than a preset speed. That is, the theft prevention control may start when the stolen vehicle is not in a high-speed driving state but in a low-speed driving state or a stopped state.
(84) FIG. **5** is a sequence diagram for describing a process of performing theft prevention release according to a theft prevention release request from a vehicle owner in the electronic parking brake system according to the embodiment.
(85) Referring to FIG. **5**, when the vehicle owner **320** wants the theft prevention release of the vehicle, the vehicle owner **320** transmits the theft prevention release request to the vehicle system **310** using the user terminal (**500**).
(86) When the vehicle system **310** receives the theft prevention release request from the vehicle owner **320**, the vehicle system **310** transmits the theft prevention release request to the electronic parking brake system **300** (**502**).
(87) When the electronic parking brake system **300** receives the theft prevention release request from the vehicle system **310**, the electronic parking brake system **300** releases the theft prevention control (**504**). During the theft prevention release, the driver is notified that the theft prevention release is operating (**504***a*). When the vehicle is driving, the operation of the electronic parking brake **10** is stopped to release the deceleration control (**504***b*). When the vehicle is stopped, a release of the forcibly engaged electronic parking brake **10** is permitted (**504***c*). The release of the

electronic parking brake **10** is switched from a prohibited state to a permitted state, and the driver may operate the EPB switch **20** to perform the parking release of the electronic parking brake **10**.
(88) After the theft prevention release is completed, the electronic parking brake system **300** transmits a theft prevention release completion notification to the vehicle system **310** (**506**).
(89) When the theft prevention control release completion notification is received from the electronic parking brake system **300**, the vehicle system **310** provides the vehicle owner **320** with the theft prevention control release completion notification (**508**).
(90) As described above, when the theft prevention request is received, the present disclosure can safely and reliably stop the stolen vehicle by generating a braking force for stop control in accordance with a driving state of the stolen vehicle.
(91) Meanwhile, the above-described controller and/or a component thereof may include one or more processor(s)/microprocessor(s) combined with a computer-readable recording medium storing a computer-readable code/algorithm/software. The processor(s)/microprocessor(s) may execute the computer-readable code/algorithm/software stored in the computer-readable recording medium to perform the above-described functions and operations.
(92) The above-described controller and/or the component thereof may further include a memory implemented as a computer-readable non-transitory recording medium or a computer-readable transitory recording medium. The memory may be controlled by the above-described controller and/or the component thereof and may be configured to store data transmitted to the above-described controller and/or the component thereof or received therefrom or store data that has been or will be processed by the above-described controller and/or the component thereof.
(93) The disclosed embodiment can also be implemented as the computer-readable code/algorithm/software on the computer-readable recording medium. The computer-readable recording medium may be the computer-readable non-transitory recording medium, such as a data storage device, which can store the data readable by the processor/microprocessor. Examples of the computer-readable recording medium include a hard disk drive (HDD), a solid state drive (SSD), a silicon disk drive (SDD), a read-only memory (ROM), a compact disc read-only memory (CD-ROM), a magnetic tape, a floppy disk, an optical data storage device, and the like.
(94) Meanwhile, although a motor-on-caliper type electronic parking brake has been described in the above embodiments, the present disclosure is not limited thereto, and an electric drum brake in which a drum rotating together with the wheel is provided thereinside, a pair of brake shoes attached to a brake lining are expanded, and thus braking is performed may be used.
(95) Further, although the electronic parking brake system having the motor-on-caliper type electronic parking brake or the electric drum brake has been described in the above embodiments, the present disclosure is not limited thereto. The electronic parking brake system may be an electromechanical brake system that performs a parking brake function of maintaining a stopped state of the vehicle during parking in addition to a service brake function of providing the braking force in a driving situation of the vehicle.
(96) As is apparent from the above description, when a theft prevention request is received, the present disclosure can safely and reliably stop a stolen vehicle by generating a braking force for stop control in accordance with a driving state of the stolen vehicle.

Claims

1. An electronic parking brake system comprising: an electronic parking brake configured to provide a braking force to a vehicle; and a controller electrically connected to the electronic parking brake and configured to control the electronic parking brake, wherein the controller is configured to perform theft prevention control that dynamically controls the braking force of the electronic parking brake according to a driving state of a stolen vehicle by maintaining a target deceleration by controlling the electronic parking brake while the stolen vehicle is driving, and

forcibly engaging the electronic parking brake while the stolen vehicle is stopping, based on reception of a theft prevention request.
2. The electronic parking brake system of claim 1, wherein the controller is configured to perform deceleration control in which a deceleration of the stolen vehicle is maintained at the target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake, based on driving of the stolen vehicle.
3. The electronic parking brake system of claim 2, wherein the controller is configured to forcibly engage the electronic parking brake, based on stopping of the stolen vehicle by the deceleration control.
4. The electronic parking brake system of claim 1, wherein, while performing the theft prevention control, the controller is configured to: ignore a parking release request from an electronic parking brake (EPB) switch and prohibit an operation stop of the electronic parking brake or a release of the electronic parking brake.
5. The electronic parking brake system of claim 1, wherein the controller is configured to: perform deceleration control in which a deceleration of the stolen vehicle is maintained at the target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake, based on a speed of the stolen vehicle, which is higher than a preset speed, and forcibly engage the electronic parking brake, based on the speed of the stolen vehicle, which is lower than the preset speed.
6. The electronic parking brake system of claim 1, wherein the controller is configured to start the theft prevention control at a time point at which an engagement of the electronic parking brake is requested.
7. The electronic parking brake system of claim 1, wherein the controller is configured to start deceleration control in which a deceleration of the stolen vehicle is maintained at the target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake at a time point at which a speed of the stolen vehicle is lower than a preset speed.
8. The electronic parking brake system of claim 1, wherein the controller is configured to output theft prevention control completion so that a vehicle owner identifies completion of the theft prevention control, based on the completion of the theft prevention control.
9. The electronic parking brake system of claim 1, wherein the controller is configured to release the theft prevention control, based on reception of a theft prevention release request.
10. The electronic parking brake system of claim 9, further comprising: a warner configured to guide information to a driver, wherein the controller is configured to output, to the warner, information notifying that the theft prevention control is released, based on the reception of the theft prevention release request.
11. The electronic parking brake system of claim 10, wherein the controller is configured to output, to the warner, information notifying that a release of the electronic parking brake is permitted, based on forcibly engaging of the electronic parking brake by the theft prevention control.
12. A method of controlling an electronic parking brake system configured to control an electronic parking brake configured to provide a braking force to a vehicle, the method comprising: determining whether a theft prevention request is received; determining a driving state of a stolen vehicle, based on the reception of the theft prevention request; and performing theft prevention control that dynamically controls the braking force of the electronic parking brake according to the driving state of the stolen vehicle, based on reception of a theft prevention request, wherein the performing of the theft prevention control comprises: performing deceleration control in which a deceleration of the stolen vehicle is maintained at a target deceleration until the stolen vehicle is stopped by engaging the electronic parking brake, based on driving of the stolen vehicle, and forcibly engaging the electronic parking brake, based on stopping of the stolen vehicle by the deceleration control.
13. The method of claim 12, wherein the performing of the theft prevention control comprises: ignoring a parking release request from an electronic parking brake (EPB) switch while the theft

prevention control is performed and prohibiting an operation stop of the electronic parking brake or a release of the electronic parking brake.

14. The method of claim 12, wherein the performing of the theft prevention control comprises starting the theft prevention control at a time point at which an engagement of the electronic barking brake is requested or at a time point at which a speed of the stolen vehicle is lower than a preset speed.

15. The method of claim 12, further comprising: outputting theft prevention control completion so that a vehicle owner identifies completion of the theft prevention control, based on the completion of the theft prevention control.

16. The method of claim 12, further comprising: releasing the theft prevention control, based on reception of a theft prevention release request, and outputting information notifying that a release of the electronic parking brake is permitted, based on forcible engaging of the electronic parking brake by the theft prevention control.